PROJECT PROPOSAL

AL-2002 Artificial Intelligence (Laboratory) | Fall 2026 | 1 mark

Instructor: Miss Amma Aslam **Section:** BCY-5A **Submission date:** 17 October 2026

Group Information

Position	Name	Roll number	Section	Primary responsibility
Leader	Muhammad Hasnain Memon	24K-2001	BCY-5A	Data pipeline & integration
Member 2	Akbar Hussain	24K-2026	BCY-5A	Classifiers & explanations
Member 3	Ateeb Ahmed Qureshi	24K-2036	BCY-5A	Clustering, EDA & UI

Project title	Explainable Fake News Headline Analyzer
Project idea	Idea No. 19 from the official list [X] Own idea [ ]
Category	Part B - Applied / Data [X] Part A - Game [ ]
GitHub repository URL	Pending - insert public repository URL before submission

Administrative note: The supplied proposal template includes a third-member row, but the separate project guidelines state a maximum of two members. This three-member group requires explicit approval from Miss Amma Aslam.

1. Project Title and Summary

This project proposes an interactive application that classifies English news headlines as likely real or likely fake using machine learning. Users will enter a headline in a simple web interface and receive a prediction, an estimated confidence score and the words that most influenced the result. The system will compare Naive Bayes and Logistic Regression using TF-IDF text features, while K-Means clustering will explore common headline topics. Unlike fixed keyword rules, trained models learn patterns from labelled examples and can evaluate new combinations of words. The application is intended as an educational decision-support demonstration, not a substitute for independent fact-checking.

2. Problem Statement and Motivation

Misleading headlines can circulate quickly, while readers usually have limited time to check each claim against trusted reporting. A headline is short and often ambiguous: the same words can appear in genuine reports, satire, opinion or fabricated content. A fixed list of suspicious phrases therefore fails to capture the range of possible writing styles. Even a moderate vocabulary of 10,000 terms creates an enormous number of possible word combinations, making manual rule-writing or brute-force checking impractical. Students, online readers and content reviewers face this uncertainty and often rely on intuition or time-consuming external searches. We will frame the task as supervised binary classification, learning statistical associations between headline text and labels in an existing dataset. We will also inspect clusters of related headlines and expose influential words, so results are interpretable. Because a headline alone cannot establish truth, predictions will be presented as model estimates rather than verified facts. The project is worthwhile because it demonstrates explainable AI and careful evaluation on a socially relevant problem.

3. Problem Formulation

3.1 PEAS Description

Performance measure	Macro-F1, class-wise precision and recall, accuracy, false positives/negatives, and reproducibility.
Environment	Offline English-headline dataset during training; user-submitted headline and saved model in the web application.
Actuators	Display a label, estimated class probabilities, influential words, topic grouping and charts.
Sensors	Headline text entered by the user; headline and reference label in training/test records.

3.2 Environment and Agent Type

Property	Selected classification	Reason
Observability	Partially observable	The headline is visible; the underlying factual truth is not.
Determinism	Stochastic	Observed labels and language patterns are uncertain.
Change over time	Static	The text does not change while one prediction is processed.
State and actions	Discrete	Tokens, classes and displayed actions are discrete.
Agents	Single-agent	One classifier makes each prediction.
Agent type	Learning	Model parameters are learned from labelled examples.

3.3 Problem Representation

State representation	Cleaned headline plus sparse TF-IDF feature vector.
Actions / moves	Predict real or fake; return confidence estimate and feature contributions.
Initial state	A text headline with no predicted label (or labelled input in training).
Goal or terminal test	Output a binary label; measure it against held-out dataset labels.
Cost / evaluation	Macro-F1 as primary metric; precision, recall and confusion matrix to inspect errors.

4. Proposed AI Approach

#	Technique	Role in the project	Implementation
1	TF-IDF + Naive Bayes	Baseline supervised text classifier	scikit-learn; own pipeline
2	TF-IDF + Logistic Regression	Second supervised classifier; word-level weights	scikit-learn; own pipeline
3	K-Means clustering	Unsupervised grouping of headline topics	scikit-learn; own interpretation
B	Majority-class dummy	Simple comparison baseline	scikit-learn DummyClassifier

TF-IDF converts the observed headline into a numerical representation of informative words, which suits short, sparse text. Multinomial Naive Bayes gives an easy-to-explain probabilistic classifier using word evidence, while Logistic Regression provides a contrasting linear decision model. Both will be trained and tested under identical splits so that their results are comparable. For explanations, we will display the highest contributing TF-IDF word weights from the fitted linear model, identifying the direction of the prediction rather than claiming causation. K-Means will group similar headlines without using class labels, and top cluster terms will help interpret possible topics. A majority-class classifier provides a transparent lower benchmark. The main evaluation criteria will be macro-F1, precision, recall and confusion matrices, with accuracy as a supplementary measure. All preprocessing and model settings will be recorded for repeatable experiments.

4.1 Planned Class Design

NewsDataset - loads, validates, de-duplicates and partitions headlines. **HeadlineAgent (abstract)** - common fit/predict interface. **NaiveBayesAgent** and **LogisticAgent** - inherit and override fit/predict. **TopicClusterer** - fits K-Means and returns topic terms. **ExperimentRunner** - performs consistent comparisons and produces charts.

5. Scope

#	In-scope feature	Why it is essential
1	Cleaned labelled headlines + EDA	Quantifies distributions, lengths, words and sources.
2	TF-IDF + Naive Bayes vs Logistic Regression	Implements and compares two learning models.
3	Explainable predictions	Highlights model-influential words alongside predictions.
4	K-Means topic discovery	Groups unlabelled headline representations.
5	Streamlit application + results dashboard	Interactive input, metrics, clusters and visual evidence.

Out of scope: Real-time news scraping, live online fact-checking, article-body verification, non-English classification, transformer/deep-learning systems, paid APIs and claims of guaranteed truth detection.

6. Data or Environment Source

Source name and URL	ISOT Fake News Dataset (University of Victoria); public Kaggle mirror: https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset
Type	Public dataset [X] Web API [ ] Self-collected [ ] Generated [ ]
Access requirements	Free Kaggle account may be required for manual download; a free API token is needed for scripted Kaggle downloads. No paid key.
Approximate size	About 45,000 English news articles, split across Fake.csv and True.csv; use the title field only.
Licence / terms of use	Kaggle listing: CC BY-NC-SA 4.0. Preserve attribution; educational, non-commercial use only; verify before publishing a subset.
Offline fallback	Cache a licence-compliant labelled subset (target: 1,000 headlines) and fitted demo models locally. No network call during normal use; ignore full raw dataset in Git.

7. Evaluation Plan

What will be compared	TF-IDF + Multinomial Naive Bayes vs TF-IDF + Logistic Regression vs majority-class baseline.
Metric(s)	Primary: macro-F1. Secondary: accuracy, per-class precision/recall, confusion matrix and training time.
Baseline	DummyClassifier(strategy="most_frequent") trained on the same training split.
Test set / instances	De-duplicate headlines before stratified 80/20 train/test split; keep test data untouched through model choice.
Runs and seed	Stratified 5-fold cross-validation within training set; fixed shuffle seed 42; one final held-out test evaluation.
Planned charts (>=3)	(1) EDA: headline-length distributions; (2) model macro-F1 comparison with cross-validation spread; (3) test confusion matrices; (4) K-Means cluster sizes/top terms.

Evaluation safeguards: Fit TF-IDF separately inside each training fold to prevent test leakage. Inspect source/class imbalance and report domain-generalization limitations; do not fabricate results.

8. Division of Work

Member	Modules / components owned	Approx. share
Muhammad Hasnain Memon (leader)	NewsDataset, cleaning, TF-IDF pipeline, GitHub integration, reproducible tests and README.	34%
Akbar Hussain	NaiveBayesAgent, LogisticAgent, baseline, model evaluation, keyword explanations.	33%
Ateeb Ahmed Qureshi	EDA, TopicClusterer, Streamlit pages, plotting and interface test cases.	33%

Each member will commit their own work, document their contributions and explain their modules in the individual viva.

9. Milestones and Timeline

#	Demonstrable milestone	Target	Owner
1	Repository ready; dataset validated; duplicate-free EDA summary produced.	Week 9	Hasnain + Ateeb
2	TF-IDF, both classifiers and majority baseline train and predict.	Week 10	Akbar + Hasnain
3	Cross-validation, held-out metrics and explanatory word contributions verified.	Weeks 11-12	Akbar
4	K-Means topics and complete Streamlit interface operate offline.	Weeks 13-14	Ateeb
5	Three+ labelled charts, clean-clone test, PDF/MD report and final repository submitted.	Week 15	All members

Specific calendar dates will be aligned with the lab instructor's announced schedule; final repository submission is due in Week 15.

10. Expected Outcome and Risks

The final submission will launch a local Streamlit dashboard where a user types a headline, views a likely real/fake prediction, and sees the words contributing to that prediction. A second view will present exploratory statistics, K-Means topics and comparison charts for the two supervised classifiers and baseline. We will report measured performance on an untouched test set rather than promise a particular accuracy. Risk 1: labels and publisher-specific wording may bias a headline-only model; we will remove duplicates, examine class/source patterns and explain limits of generalization. Risk 2: downloading or processing the full dataset may be inconvenient; a documented, licence-compliant local sample and saved models will keep the application usable offline.

For Office Use - Lab Instructor

Instructor	Miss Amma Aslam
Proposal status	[ ] Approved [ ] Approved with changes [ ] Revision required
Comments	_____
Signature and date	_____

Academic integrity: The team will attribute the dataset, cited references and any AI assistance, and can explain the submitted implementation.